

MCHA4400

Vision-based navigation

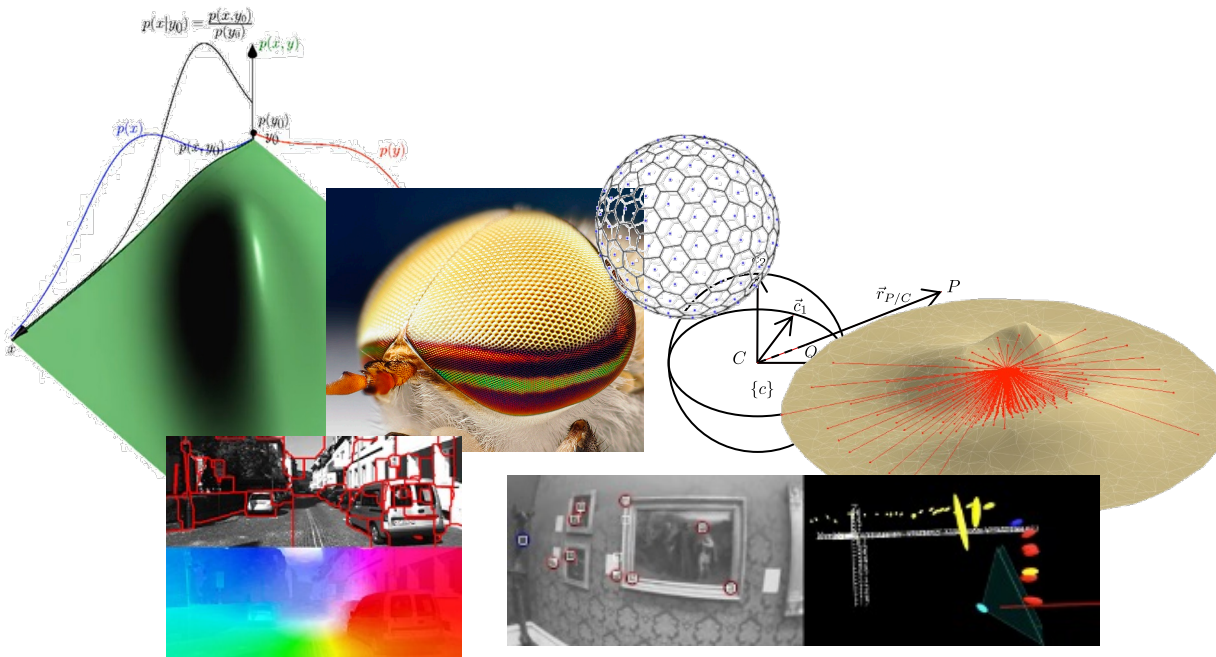


THE UNIVERSITY OF
NEWCASTLE
AUSTRALIA

FACULTY OF
ENGINEERING AND
BUILT ENVIRONMENT



www.newcastle.edu.au



Dr Chris Renton
School of Engineering

Assignment 2

Scenario 4: outdoor navigation



Scenario 4: outdoor navigation

Subtitle file (flight.SRT) format:

98

00:00:01,618 --> 00:00:01,634

FrameCnt : 98, DiffTime : 16ms

2021-09-29 11:13:39,740,203

[iso : 110] [shutter : 1/400.0] [fnum : 280] [ev : 0] [ct : 6000] [color_md : default] [focal_len : 280] [latitude : -32.868821] [longitude : 151.684684] [**altitude: 62.221001**]

99

00:00:01,634 --> 00:00:01,651

FrameCnt : 99, DiffTime : 17ms

2021-09-29 11:13:39,756,888

[iso : 110] [shutter : 1/400.0] [fnum : 280] [ev : 0] [ct : 6000] [color_md : default] [focal_len : 280] [latitude : -32.868819] [longitude : 151.684679] [**altitude: 62.185001**]

100

00:00:01,651 --> 00:00:01,668

FrameCnt : 100, DiffTime : 17ms

2021-09-29 11:13:39,773,571

[iso : 110] [shutter : 1/400.0] [fnum : 280] [ev : 0] [ct : 6000] [color_md : default] [focal_len : 280] [latitude : -32.868819] [longitude : 151.684679] [**altitude: 62.185001**]

Scenario 5: identical tag landmarks

5



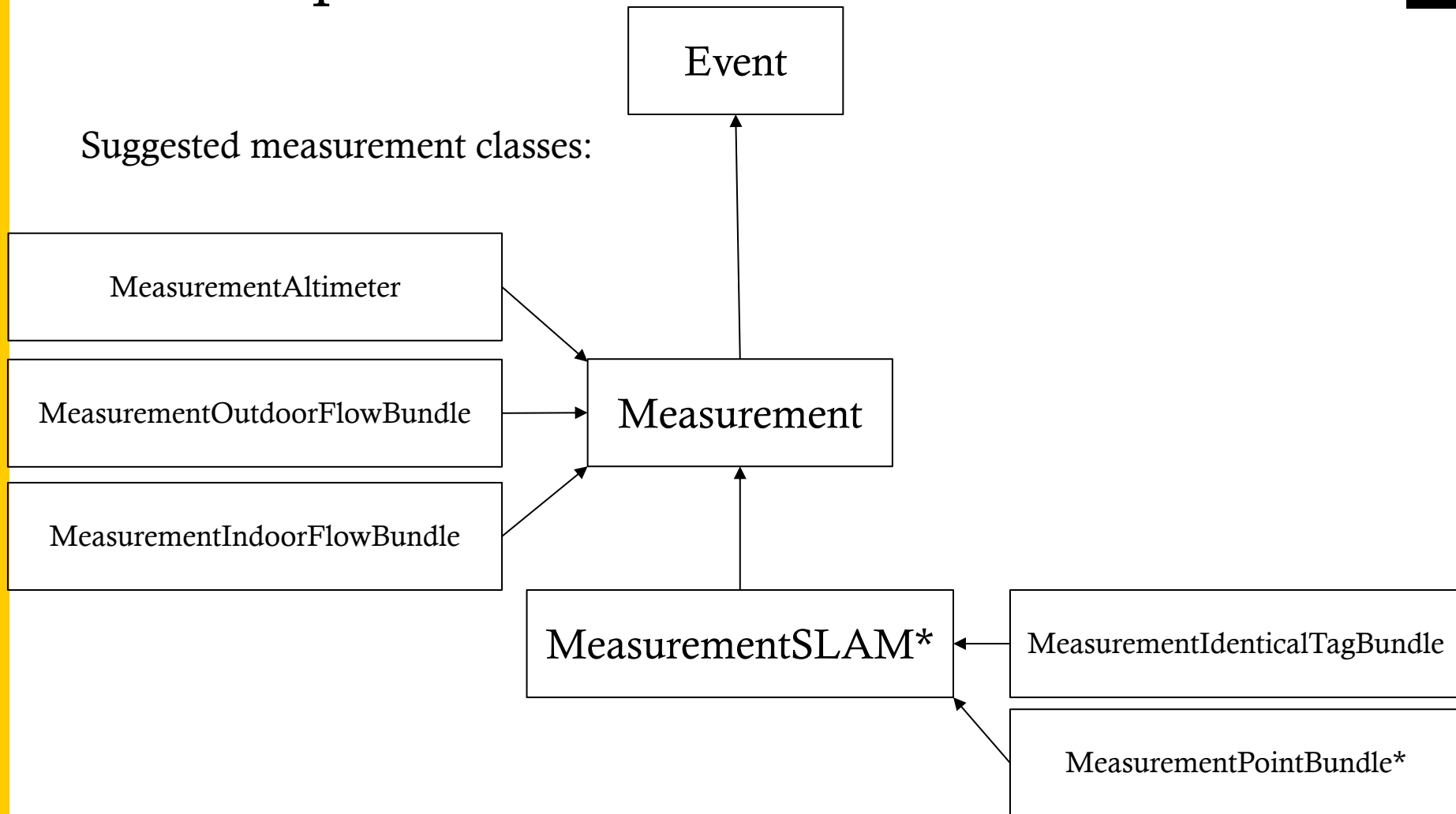
Scenario 6: indoor navigation

6



Some tips

Suggested measurement classes:



*Modify the landmark index offset from 12 to 18.

Some tips

Event-based sequential measurement updates

Scenario 4:

```
MeasurementAltimeter measAlt(t, ...);  
MeasurementOutdoorFlowBundle measFlow(t, ...);  
...  
measAlt.process(system); // Includes time update from previous frame  
measFlow.process(system); // No time update, since no time elapsed since previous event
```

Scenario 5:

```
MeasurementIndoorFlowBundle measFlow(t, ...);  
MeasurementIdenticalTagBundle measTag(t, ...);  
...  
measFlow.process(system); // Includes time update from previous frame  
measTag.process(system); // No time update, since no time elapsed since previous event
```

Scenario 6:

```
MeasurementIndoorFlowBundle measFlow(t, ...);  
MeasurementPointBundle measPoint(t, ...);  
...  
measFlow.process(system); // Includes time update from previous frame  
measPoint.process(system); // No time update, since no time elapsed since previous event
```


Some tips

To implement the map management in Scenario 6, the `MeasurementPointBundle` class can override the `update()` function.

Declare in `MeasurementPointBundle.h`:

```
virtual void update(SystemEstimator & system) override;
```

Implement in `MeasurementPointBundle.cpp`:

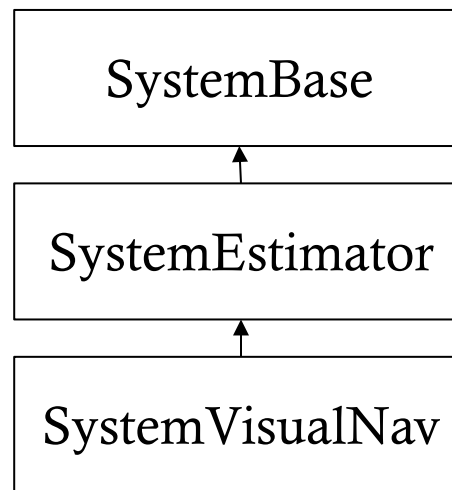
```
void MeasurementPointBundle::update(SystemEstimator & system_) override
{
    SystemVisualNav & system = dynamic_cast<SystemVisualNav &>(system_);
    // Identify landmarks with matching features (data association)
    // Remove failed landmarks from map (consecutive failures to match)
    // Identify surplus features that do not correspond to landmarks in the map
    // Initialise up to Nmax - N new landmarks from best surplus features

    Measurement::update(system); // Do the actual measurement update
}
```

Similarly for Scenario 5 with `MeasurementIdenticalTagBundle::update`.

Some tips

Suggested system classes:



Assume point-like landmarks for the map partition.

Scenario 4: No landmarks need to be initialised.

Scenario 5: Landmarks initialised with `MeasurementIdenticalTagBundle::update()`

Scenario 6: Landmarks initialised with `MeasurementPointBundle::update()`

Stochastic cloning

The time update requires the following function

$$\Phi: \begin{bmatrix} \mathbf{x}[k] \\ \Delta \mathbf{w}_\nu[k] \end{bmatrix} \mapsto \mathbf{x}[k+1] \quad \Delta \mathbf{w}_\nu[k] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_\nu \Delta t)$$

For 2-frame optical flow with landmarks, this function becomes

$$\Phi: \begin{bmatrix} \nu[k] \\ \eta[k] \\ \zeta[k] \\ \mathbf{m}[k] \\ \Delta \mathbf{w}_\nu[k] \end{bmatrix} \mapsto \begin{bmatrix} \nu[k+1] \\ \eta[k+1] \\ \zeta[k+1] \\ \mathbf{m}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{z}[k+1] \\ \zeta[k+1] \\ \mathbf{m}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{f}_k(\mathbf{z}[k], \Delta \mathbf{w}_\nu[k]) \\ \eta[k] \\ \mathbf{m}[k] \end{bmatrix}$$

where we only need to integrate the following SDE to obtain $\mathbf{z}[k+1]$:

$$\underbrace{\begin{bmatrix} \dot{\nu}(t) \\ \dot{\eta}(t) \end{bmatrix}}_{\dot{\mathbf{z}}(t)} = \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{J}_K(\eta(t))\nu(t) \end{bmatrix}}_{\mathbf{f}(\mathbf{z}(t))} + \underbrace{\begin{bmatrix} \dot{\mathbf{w}}_\nu(t) \\ \mathbf{0} \end{bmatrix}}_{\dot{\mathbf{w}}(t)} \quad \dot{\mathbf{w}}_\nu(t) \sim \mathcal{GP}(\mathbf{0}, \mathbf{Q}_\nu \delta(t-t'))$$

Stochastic cloning

Since ζ is the pose at the previous flow measurement and we have other events that should not modify the ζ state, we can modify the Φ function as follows:

$$\Phi: \begin{bmatrix} \nu[k] \\ \eta[k] \\ \zeta[k] \\ \mathbf{m}[k] \\ \Delta \mathbf{w}_\nu[k] \end{bmatrix} \mapsto \begin{bmatrix} \nu[k+1] \\ \eta[k+1] \\ \zeta[k+1] \\ \mathbf{m}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{z}[k+1] \\ \zeta[k+1] \\ \mathbf{m}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{f}_k(\mathbf{z}[k], \Delta \mathbf{w}_\nu[k]) \\ \xi[k] \\ \mathbf{m}[k] \end{bmatrix}$$

where we define $\xi[k]$ as

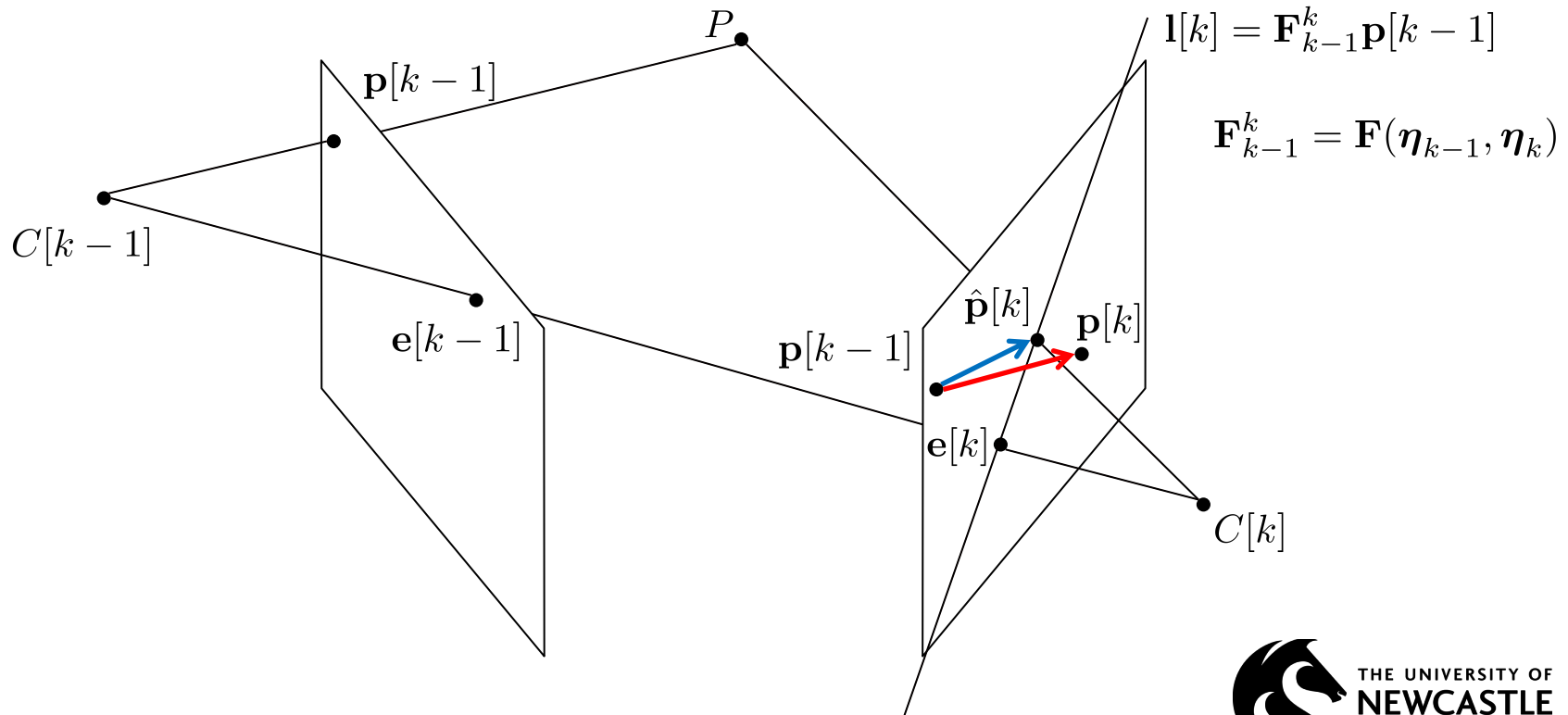
$$\zeta[k+1] = \xi[k] \triangleq \begin{cases} \eta[k] & \text{if handling flow measurement event} \\ \zeta[k] & \text{if handling any other event} \end{cases}$$

This effectively holds the state of ζ until the next flow event.

The implementation amounts to selecting a different offset into the state vector when implementing the Φ function.

Epipolar flow prediction

In scenarios 5 & 6, you are asked to plot the predicted flow field using a flow measurement model based on the epipolar constraint. To do this, you can project the measured points in frame k onto the epipolar line computed from frame $k-1$.



Epipolar flow prediction

To project the measured points in frame k onto the epipolar line computed from frame $k-1$, we can consider the following constrained optimisation problem:

$$\begin{aligned} \hat{\mathbf{p}}[k] = \arg \min_{\hat{\mathbf{p}}} & \left\| \mathbf{r}(\hat{\mathbf{p}}) - \mathbf{r}(\mathbf{p}[k]) \right\| \\ \text{subject to} & \mathbf{l}[k]^T \hat{\mathbf{p}} = 0 \end{aligned} \quad \mathbf{r}: \begin{bmatrix} u \\ v \\ w \end{bmatrix} \mapsto \begin{bmatrix} \frac{u}{w} \\ \frac{v}{w} \end{bmatrix}$$

Due to the nonlinearity of the map $\mathbf{r}: \mathbb{P}^2 \rightarrow \mathbb{R}^2$, we can pose an equivalent problem in normalised homogeneous coordinates,

$$\begin{aligned} \hat{\mathbf{p}}[k] = \arg \min_{\hat{\mathbf{p}}} & \left\| \hat{\mathbf{p}} - \mathbf{n}_p(\mathbf{p}[k]) \right\| \\ \text{subject to} & \mathbf{l}[k]^T \hat{\mathbf{p}} = 0 \text{ and } \mathbf{e}_3^T \hat{\mathbf{p}} = 1 \end{aligned} \quad \mathbf{n}_p: \begin{bmatrix} u \\ v \\ w \end{bmatrix} \mapsto \begin{bmatrix} \frac{u}{w} \\ \frac{v}{w} \\ 1 \end{bmatrix}$$

This is a least norm problem subject to linear inequality constraints.

Constrained least norm

Consider the following constrained optimisation problem:

$$\begin{aligned} \mathbf{x}^* &= \arg \min_{\mathbf{x}} \|\mathbf{x} - \mathbf{x}_0\|, \\ &\text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is full row rank.

Then, let

$$\mathbf{A}^\top = \underbrace{[\mathbf{Y} \quad \mathbf{Z}]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_1 \\ \mathbf{0}_{(n-m) \times m} \end{bmatrix}}_{\mathbf{R}} = \mathbf{Y}\mathbf{R}_1$$

be a QR decomposition.

Substituting for the thin decomposition of $\mathbf{A}$, the constraint may be rewritten as

$$\mathbf{R}_1^\top \mathbf{Y}^\top \mathbf{x} = \mathbf{b}$$

Constrained least norm

Let $\mathbf{x}$ be written as combination of range and null space components,

$$\mathbf{x} = \mathbf{Y}\mathbf{x}_Y + \mathbf{Z}\mathbf{x}_Z$$

Substituting this into the constraint equation yields

$$\mathbf{R}_1^T \mathbf{Y}^T (\mathbf{Y}\mathbf{x}_Y + \mathbf{Z}\mathbf{x}_Z) = \mathbf{b}$$

$$\mathbf{R}_1^T \mathbf{x}_Y = \mathbf{b}$$

$$\mathbf{x}_Y = \mathbf{R}_1^{-T} \mathbf{b}$$

and therefore $\mathbf{x} = \mathbf{Y}\mathbf{R}_1^{-T} \mathbf{b} + \mathbf{Z}\mathbf{x}_Z$

To find $\mathbf{x}_Z$ that minimises the norm, note that this is equivalent to minimising the square of the norm,

$$\begin{aligned} \|\mathbf{x} - \mathbf{x}_0\|^2 &= \|\mathbf{Z}\mathbf{x}_Z + \mathbf{Y}\mathbf{R}_1^{-T} \mathbf{b} - \mathbf{x}_0\|^2 \\ &= \mathbf{x}_Z^T \mathbf{Z}^T \mathbf{Z} \mathbf{x}_Z + 2\mathbf{x}_Z^T \mathbf{Z}^T (\mathbf{Y}\mathbf{R}_1^{-T} \mathbf{b} - \mathbf{x}_0) + \text{const} \\ &= \mathbf{x}_Z^T \mathbf{x}_Z - 2\mathbf{x}_Z^T \mathbf{Z}^T \mathbf{x}_0 + \text{const} \end{aligned}$$

which is minimised for $\mathbf{x}_Z = \mathbf{Z}^T \mathbf{x}_0$

Constrained least norm

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{x}_0 \in \mathbb{R}^n$ where $m < n$ and $\mathbf{A}$ is full row rank. Then, the solution to the constrained least norm problem

$$\begin{aligned} \mathbf{x}^* &= \arg \min_{\mathbf{x}} \|\mathbf{x} - \mathbf{x}_0\|, \\ &\text{subject to } \mathbf{Ax} = \mathbf{b}, \end{aligned}$$

is given by

$$\mathbf{x}^* = \mathbf{Y}\mathbf{R}_1^{-\top} \mathbf{b} + \mathbf{Z}\mathbf{Z}^\top \mathbf{x}_0$$

where

$$\mathbf{A}^\top = \underbrace{\begin{bmatrix} \mathbf{Y} & \mathbf{Z} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_1 \\ \mathbf{0}_{(n-m) \times m} \end{bmatrix}}_{\mathbf{R}} = \mathbf{Y}\mathbf{R}_1$$

is a QR decomposition where $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is an orthogonal matrix with column partitions $\mathbf{Y} \in \mathbb{R}^{n \times m}$ and $\mathbf{Z} \in \mathbb{R}^{n \times (n-m)}$ and $\mathbf{R} \in \mathbb{R}^{n \times m}$ is an upper triangular matrix with upper partition $\mathbf{R}_1 \in \mathbb{R}^{m \times m}$.

Epipolar flow prediction

To project the measured points in frame k onto the epipolar line computed from frame $k-1$, we can consider the following constrained optimisation problem:

$$\begin{aligned} \hat{\mathbf{p}}[k] &= \arg \min_{\hat{\mathbf{p}}} \|\hat{\mathbf{p}} - \mathbf{n}_p(\mathbf{p}[k])\| \\ \text{subject to } \begin{bmatrix} \mathbf{l}[k]^\top \\ \mathbf{e}_3^\top \end{bmatrix} \hat{\mathbf{p}} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$\mathbf{n}_p: \begin{bmatrix} u \\ v \\ w \end{bmatrix} \mapsto \begin{bmatrix} \frac{u}{w} \\ \frac{v}{w} \\ 1 \end{bmatrix}$$

$$\mathbf{l}[k] = \mathbf{F}(\boldsymbol{\eta}_{k-1}, \boldsymbol{\eta}_k) \mathbf{p}[k-1]$$

Applying the constrained least norm lemma yields

$$\hat{\mathbf{p}}[k] = \mathbf{Y}\mathbf{R}_1^{-\top} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \mathbf{Z}\mathbf{Z}^\top \mathbf{n}_p(\mathbf{p}[k])$$

where

$$\begin{bmatrix} \mathbf{l}[k] & \mathbf{e}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{Y} & \mathbf{Z} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_1 \\ \mathbf{0}_{1 \times 2} \end{bmatrix}}_{\mathbf{R}}$$

is a full QR decomposition.

Then you can reapply the lens distortion and draw the predicted flow vectors on the image frame.